

RateIT — Hackathon Phase 1 Submission

One platform to rate everything. Trusted reviews. No fakes.

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1. The Problem We Are Solving

Today, if you want honest reviews, you face three big problems:

Problem 1: Too Many Platforms, No Single Source of Truth

Want to rate a coaching class? There is no good platform for that. Want to rate a PG hostel? You are stuck with random Facebook groups. Want to rate a local café? You go to Google Maps. Every category lives on a different platform. Users are scattered, and no single place collects all this feedback in a reliable way.

Problem 2: Fake Reviews Are Everywhere

Business owners buy fake 5-star reviews. Competitors post fake 1-star reviews. Bots copy-paste the same text across dozens of places. The result? Nobody trusts online ratings anymore. A 2024 study found that nearly 42% of online reviews are suspected to be fake or incentivized.

Problem 3: Students and Young Professionals Are Underserved

Platforms like Zomato cover restaurants, but nobody covers the things that actually matter to students — coaching centers, PG hostels, local repair shops, college canteens. These are real everyday decisions with no reliable review data.

In short: Reviews are broken. They are fragmented, fake, and missing for the categories that matter most.

Engineering Goals

Every design decision in RatelT maps to one of these explicit engineering objectives:

- 1. **Zero-Trust Input Architecture** — Never trust the client. Every review payload is re-validated server-side with Zod schemas, regardless of what the frontend already checked. No raw user input ever reaches a SQL query.
- 2. **Sub-50ms Synchronous Fraud Scoring** — The entire fraud heuristics pipeline (Jaccard + velocity + burst check) must complete in under 50ms so the user does not perceive a delay on review submission. We achieve this by using O(n) set operations instead of expensive NLP.
- 3. **Client-Side EXIF Pre-Check** — Photo metadata is extracted in the browser using `exifr` (dynamic import to avoid bloating the main bundle). Only the extracted GPS/timestamp/device fields are sent to the server — the raw photo never leaves the client for EXIF purposes.
- 4. **Graceful Degradation of Optional Services** — If Upstash Redis is unreachable, rate limiting is silently skipped (reviews still work). If EXIF extraction fails, the review proceeds without a photo bonus. No single optional service can break the core submission flow.
- 5. **One Review Per User Per Place (Database-Enforced)** — The `UNIQUE(userId, placeId)` constraint is enforced at the PostgreSQL level, not just in application code. Even if the API has a race condition, the database rejects duplicate reviews.
- 6. **Weighted Aggregate Ratings** — A place's displayed rating is not a simple average. Each review's contribution is multiplied by the reviewer's `trustWeight` (derived from their trust tier), so a Trusted reviewer's 4-star counts 5x more than a Suspicious reviewer's 4-star.

2. Our Idea — What Is RatelT?

RatelT is a web platform where anyone can rate and review **anything** — coaching classes, PG hostels, cafés, restaurants, local services — all in one place.

What makes us different is **trust**. Every review goes through a built-in anti-fraud system that catches fake reviews, bot spam, and suspicious behavior **before** a review is published. No manual moderation needed. No expensive AI. Just smart math.

The core concept in one line:

"Rate anything. Trust every review. One platform."

What RatelT Offers:

For Users	For Businesses
Rate any place with 1-5 stars	Claim your business listing
Write reviews with photos	Reply to customer reviews
GPS check-in to prove you were there	See aggregate feedback and trends
Save favorites into personal lists	Earn a "Verified Business" badge
Earn trust badges for honest reviewing	Reach customers who actually visited
Browse categories that matter (Coaching, PGs, Cafés)	Get rated fairly — fake reviews are auto-detected

Who Is It For?

- **Students** looking for reliable coaching classes and PGs in cities like Pune
- **Young professionals** discovering local cafés and services

- **Business owners** who want honest feedback and a chance to respond
 - **Anyone** making everyday decisions based on other people's experiences
-

3. How We Will Build It — Our Approach

Our approach is simple: build a working product with real fraud detection from day one. Not just a pretty UI — a functioning system that actually catches fakes.

Architecture in Plain English:

1. **The user opens the website** → Next.js serves the page instantly (server-side rendered for speed and SEO)
2. **The user searches for a place** → The query hits our PostgreSQL database through Prisma ORM, with indexed columns for fast results
3. **The user writes a review** → Before saving, the review passes through our **Fraud Heuristics Engine** which checks for:
 - Copy-pasted text (Jaccard similarity on word bigrams)
 - Bulk posting behavior (too many reviews in a short window)
 - New account spam (brand new accounts flooding reviews)
 - GPS spoofing (impossible travel between reviewed locations)
 - Fake photos (EXIF metadata cross-check against GPS)
4. **If the review is clean** → It is saved to the database and the place's average rating is recalculated
5. **If the review is suspicious** → It is shadow-hidden and the user's trust score is penalized

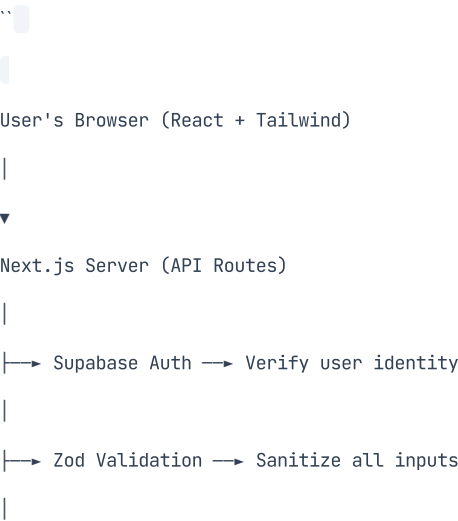
This is not a prototype concept — every piece described above is **already implemented** in our codebase.

4. Tools and Technology We Use

What	Tool	Why We Chose It
Frontend	Next.js 14 (App Router) + React 18	Server-side rendering for fast page loads and SEO. App Router for modern file-based routing.
Styling	Tailwind CSS + shadcn/ui	Rapid development with consistent design. shadcn gives us accessible, production-ready UI components.
Database	PostgreSQL (via Supabase)	Relational database for complex queries (user → reviews → places). Supabase provides managed hosting + real-time features.
ORM	Prisma	Type-safe database queries in TypeScript. Auto-generates client from our schema. Zero raw SQL needed.
Authentication	Supabase Auth	Phone OTP and email login out of the box. No need to build auth from scratch.
Rate Limiting	Upstash Redis + @upstash/ratelimit	Sliding window rate limiting to prevent API abuse. 5 reviews per 10 minutes, 10 OTP requests per hour.
Form Validation	Zod	Runtime schema validation for all API inputs — ratings (1-5), text (max 500 chars), UUIDs, GPS coordinates.
Photo Processing	exifr	Extracts GPS, timestamp, and device info from uploaded photos for fraud cross-checking.
Geolocation	Browser Geolocation API + Haversine formula	Verifies the reviewer is physically near the place (within 50m geofence).
CAPTCHA	Cloudflare Turnstile (@marsidev/react-turnstile)	Bot protection at login and review submission without annoying image puzzles.
Testing	Playwright	End-to-end browser testing for authentication, navigation, lists, and error handling flows.
Charts	Recharts	Dashboard visualizations for user analytics and rating trends.
Deployment	Vercel	Zero-config deployment for Next.js with global CDN and automatic HTTPS.
Smooth Scrolling	Lenis (@studio-freight/lenis)	Butter-smooth scroll animations for a premium feel.

5. System Architecture and Data Flow

How data flows through the system:



└─> Upstash Rate Limiter ─> Block abuse (5 reviews / 10 min)

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└─> Fraud Heuristics Engine ─> Score review risk (0-100)

| └─ Jaccard text similarity

| └─ Bulk posting velocity check

| └─ New account burst detection

| └─ EXIF photo cross-check

|

└─> Trust Score Engine ─> Update user trust tier

| └─ Phone verification: +10

| └─ Account age: +5/month (max +15)

| └─ Check-in verified reviews: +10

| └─ EXIF match bonus: +5

| └─ Flag penalties: -5/flag (max -25)

| └─ Velocity penalty: -15

| └─ Impossible travel: -10

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PostgreSQL Database (via Prisma ORM)

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└─ Users (trust scores, verification status)

└─ Places (avg ratings, geofence radius, categories)

└─ Reviews (ratings, text, GPS check-in, photos)

└─ Flags (user reports on reviews)

└─ Lists (user-curated collections)

└─ BusinessClaims (owner verification requests)

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Key Design Decisions:

- **Fraud checking happens BEFORE the review is saved**, not after. This means fake data never enters our database.
- **Trust scores are computed server-side** and are never manipulable from the frontend.
- **Every API route is protected** – Supabase Auth verifies the user, then Prisma upserts their profile to ensure the database is always in sync.

6. Core Features – What the App Actually Does

6.1 Multi-Step Review Flow (Already Built)

Our review submission is not a simple form. It is a guided, multi-step experience:

Step 1: Search → User searches for a place from our database of Pune-based listings (coaching, PG, café, restaurant, local service)

Step 2: GPS Check-In → The app requests the user's location and verifies they are within 50 meters of the place using the Haversine great-circle distance formula. If verified, the review gets a "✅ Check-in Verified" badge.

Step 3: Blind Rating → User gives a 1-5 star rating BEFORE seeing other reviews. This prevents bias from existing ratings.

Step 4: Detailed Review → User writes optional text (max 500 chars), selects tags, uploads photos (with EXIF extraction), and sets visibility (Public or Private).

Step 5: Confirmation → Review summary is shown before final submission.

6.2 Category System

We focus on categories that matter to students and young professionals in Indian cities:

Category	Examples
COACHING	IIT JEE Academy, Mahesh Tutorials
PG_HOSTEL	Sunrise PG for Boys, Green Villa Hostel
CAFE	Café Good Luck
RESTAURANT	(Pune food spots)
LOCAL_SERVICE	Electricians, plumbers, tailors

6.3 User Dashboard

- **Items rated this week** – calculated from actual review timestamps
- **Favorites saved** – pulled from the user's saved lists
- **Lists created** – personal curated collections
- **Badges earned** – gamification based on activity

6.4 Business Features

- **Business Claim:** Owners can claim their listing via phone verification (auto-match) or document upload (manual review)
- **Business Reply:** Claimed owners can reply to reviews
- **Verified Badge:** Claimed and verified businesses get a visual trust indicator

6.5 Legal Pages (Already Built)

- **Privacy Policy** – fully compliant with India's DPDP Act 2023 and DPDP Rules 2025
- **Terms of Service** – user rights and platform rules
- **Grievance Redressal** – required under Indian law, with a response SLA of 24 hours
- **Moderation Policy** – explains how reviews are flagged and moderated

7. Our Anti-Fraud System – The Math Behind Trust

This is RateIT's biggest differentiator. We built two interlocking systems: **Fraud Heuristics** (catches bad reviews) and **Trust Scores** (rewards good reviewers).

7.1 Fraud Heuristics – How We Catch Fake Reviews

Every review is scored on a 0-100 risk scale. If the score reaches 25 or above, the review is automatically hidden.

Check 1: Copy-Paste Detection (Jaccard Similarity)

We break the review text into word pairs (bigrams) and compare them against the user's previous reviews.

How it works:

- Text "Great food, nice ambiance" → bigrams: {great_food, food_nice, nice_ambiance}
- Compare with previous reviews using Jaccard Index:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

- If similarity ≥ 0.60 → **Medium flag** (+15 risk)
- If similarity ≥ 0.80 → **High flag** (+30 risk)

This catches templated reviews that bots post with minor word changes.

Check 2: Bulk Posting Detection

If a user posts more than 5 reviews in 60 minutes → **High flag** (+25 risk)

Real users rarely review 5 places in an hour. Bots do.

Check 3: New Account Burst

If an account is less than 24 hours old AND has posted 3+ reviews → **Medium flag** (+20 risk)

This is the most common bot pattern: create account, flood reviews, abandon.

Check 4: EXIF Photo Cross-Check

When a user uploads a photo with their review, we extract:

- GPS coordinates from the photo
- Timestamp of when the photo was taken
- Device make and model

Then we check:

- Is the photo GPS more than 1km from the reviewed place? → Warning
- Is the photo more than 30 days old? → Warning
- Is the photo timestamp in the future? → Possible GPS spoofing

7.2 Trust Score Engine – How We Reward Honest Reviewers

Every user starts at a trust score of 50. The score goes up or down based on behavior:

Signal	Effect	Max
Phone verified	+10	+10
Account age (per month)	+5	+15
Check-in verified reviews (>80%)	+10	+10
EXIF photo matches	+2 each	+5
Flags received	-5 each	-25
Suspicious review velocity	-15	-15
Impossible travel detected	-10	-10

Trust Tiers:

Score Range	Tier	Review Weight	Badge
70-100	✔️ Trusted	1.0x (full weight)	"Trusted Reviewer"
40-69	🧑 Neutral	0.8x	"Reviewer"
1-39	⚠️ Suspicious	0.2x	"Under Review"
0	🚫 Banned	0x (reviews hidden)	"Restricted"

Why this matters: A trusted reviewer's 4-star rating counts 5x more than a suspicious reviewer's 4-star rating when calculating a place's average. This makes it very hard for fake review farms to influence overall scores.

7.3 Impossible Travel Detection (GPS Spoofing)

If a user reviews a place in Kothrud, Pune at 2:00 PM and then reviews a place in Hinjewadi (15km away) at 2:10 PM – that is physically impossible without a car going 90 km/h in city traffic. If two reviews are >50km apart within 30 minutes, we flag it.

The distance is calculated using the **Haversine formula**:

$$d = 2R \cdot \arctan2(\sqrt{a}, \sqrt{1-a})$$

where:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2\left(\frac{\Delta\lambda}{2}\right)$$

R = 6,371,000 meters (Earth's radius)

7.4 Mathematical Formulation Summary Matrix

Metric	Formula	Key Variables	Purpose	Threshold	Source File	
Text Similarity	$J(A,B) = \frac{ A \cap B }{ A \cup B }$	$A \cap B$	$ A \cup B $	$ A \cup B $		A, B = word bigram sets from review texts
Review Velocity	count(reviews) in sliding window	Window = 60 min, max = 5 reviews	Detect bot-like burst posting	> 5 reviews / 60 min	lib/fraud-heuristics.ts	
Account Burst	age < 24h AND count ≥ threshold	Account creation timestamp, review count	Catch new-account spam bots	< 24h old + ≥ 3 reviews	lib/fraud-heuristics.ts	
Composite Risk	$R = \sum(W_i \cdot F_i)$, clamped to [0, 100]	W_i = severity weight, $F_i \in \{0,1\}$ = flag indicator	Aggregate fraud signal into single score	$R \geq 25 \rightarrow$ isSuspicious	lib/fraud-heuristics.ts	
Geographic Distance	$d = 2R \cdot \arcsin(\sqrt{\frac{1}{2}(\sin^2(\frac{\varphi}{2}) + \cos(\lambda) \sin^2(\frac{\lambda}{2}))})$ (Haversine)	φ = latitude, λ = longitude, R = 6,371,000m	GPS check-in verification + impossible travel	≤ 50m (check-in), > 50km in 30min (spoofing)	lib/geolocation.ts	
Trust Score	Base(50) + $\sum(\text{signal bonuses}) - \sum(\text{penalties})$	Phone verify (+10), age (+5/mo), flags (-5 each), etc.	Compute per-reviewer credibility tier	≥ 70 Trusted, ≥ 40 Neutral, > 0 Suspicious, 0 Banned	lib/trust-score.ts	
Review Weight	$f(\text{trustScore}) \rightarrow \{1.0, 0.8, 0.5, 0.2, 0\}$	Trust score mapped to multiplier	Weight a review's contribution to aggregate rating	Trusted = 1.0x, Suspicious = 0.2x, Banned = 0x	lib/trust-score.ts	
EXIF Distance	Haversine(photo GPS, place GPS)	Photo lat/lng from EXIF, place lat/lng from DB	Verify photo was taken at the reviewed location	> 1km = warning flag	lib/exif.ts	
EXIF Freshness	reviewDate - photoDate (in days)	Photo DateTimeOriginal, review submission timestamp	Detect recycled or stock photos	> 30 days = warning, < -1 day = manipulation	lib/exif.ts	

8. Database Design – How Data Is Organized

We use PostgreSQL with 7 main tables:

Entity Relationship Diagram



| | | |

└─ List └─ Flag BizReply BusinessClaim

| |

ListItem ───────────────────────────────── Place

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Key Tables and Their Purpose:

Users – stores identity, trust score, ban status, consent tracking

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id, phone, email, displayName, trustScore (default: 50),
phoneVerified, flagCount, isBanned, consentGiven, consentDate, dataEraseReq

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Places – stores businesses and locations with geofencing

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id, name, category (COACHING | PG_HOSTEL | LOCAL_SERVICE | RESTAURANT | CAFE),
address, city (default: "Pune"), latitude, longitude,
avgRating, weightedRating, reviewCount, geofenceRadius (default: 50m),
isVerifiedBiz, claimStatus, tags, priceRange

|

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Reviews – connects users to places with full audit trail

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id, userId, placeId, rating (1-5), text, tags, photoUrls,
visibility (PUBLIC | PRIVATE), checkinLat, checkinLng, checkinVerified,
trustWeight, flagCount, isHidden, version, publishDate

|

| |

|

- Unique constraint: one review per user per place (userId + placeId)

- Indexed by placeId + publishDate for fast place pages

- Indexed by userId + createdAt for user history

Flags – user reports on suspicious reviews

id, reviewId, userId, reason (FAKE_REVIEW | OFFENSIVE | IRRELEVANT | SPAM | DEFAMATORY)

- Unique constraint: one flag per user per review

Lists / ListItems – user-curated collections of places

BusinessClaims – owner verification requests (auto via phone match or manual via document)

BizReply – business owner responses to reviews (one reply per review)

Database Indexes (for performance):

- Place(category, city) – fast category browsing

- Place(avgRating DESC) – sorted place listings

- Review(placeId, publishDate DESC) – fast review loading

- Review(flagCount) – quick moderation queries

- User(trustScore) – trust tier lookups

SQL Migration Scripts (DDL)

Below is the complete DDL derived from our Prisma schema. These are the actual SQL statements that create the database:

`sql`

```
-- =====  
  
-- ENUM TYPES  
  
-- =====
```

```

CREATE TYPE "Category" AS ENUM (
    'COACHING', 'PG_HOSTEL', 'LOCAL_SERVICE', 'RESTAURANT', 'CAFE'
);

CREATE TYPE "ClaimStatus" AS ENUM (
    'UNCLAIMED', 'PENDING', 'VERIFIED', 'REJECTED'
);

CREATE TYPE "Visibility" AS ENUM ('PUBLIC', 'PRIVATE');

CREATE TYPE "FlagReason" AS ENUM (
    'FAKE_REVIEW', 'OFFENSIVE', 'IRRELEVANT', 'SPAM', 'DEFAMATORY'
);

-- =====
-- TABLE: User
-- =====

CREATE TABLE "User" (
    "id" TEXT NOT NULL,
    "phone" TEXT,
    "email" TEXT,
    "phoneVerified" BOOLEAN NOT NULL DEFAULT false,
    "displayName" TEXT,
    "avatarUrl" TEXT,
    "trustScore" DOUBLE PRECISION NOT NULL DEFAULT 50.0,
    "accountAge" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
    "flagCount" INTEGER NOT NULL DEFAULT 0,
    "isBanned" BOOLEAN NOT NULL DEFAULT false,
    "banReason" TEXT,
    "consentGiven" BOOLEAN NOT NULL DEFAULT false,
    "consentDate" TIMESTAMP(3),
    "dataEraseReq" TIMESTAMP(3),
    "bio" TEXT,
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
    "updatedAt" TIMESTAMP(3) NOT NULL,
    CONSTRAINT "User_pkey" PRIMARY KEY ("id")
);

CREATE UNIQUE INDEX "User_phone_key" ON "User"("phone");
CREATE UNIQUE INDEX "User_email_key" ON "User"("email");
CREATE INDEX "User_phone_idx" ON "User"("phone");
CREATE INDEX "User_trustScore_idx" ON "User"("trustScore");

-- =====
-- TABLE: Place
-- =====

```

```

CREATE TABLE "Place" (
  "id" TEXT NOT NULL,
  "name" TEXT NOT NULL,
  "category" "Category" NOT NULL,
  "description" TEXT,
  "address" TEXT NOT NULL,
  "city" TEXT NOT NULL DEFAULT 'Pune',
  "latitude" DOUBLE PRECISION NOT NULL,
  "longitude" DOUBLE PRECISION NOT NULL,
  "avgRating" DOUBLE PRECISION NOT NULL DEFAULT 0,
  "weightedRating" DOUBLE PRECISION NOT NULL DEFAULT 0,
  "reviewCount" INTEGER NOT NULL DEFAULT 0,
  "geofenceRadius" INTEGER NOT NULL DEFAULT 50,
  "isVerifiedBiz" BOOLEAN NOT NULL DEFAULT false,
  "claimStatus" "ClaimStatus" NOT NULL DEFAULT 'UNCLAIMED',
  "ownerPhone" TEXT,
  "businessPhone" TEXT,
  "tags" TEXT[],
  "priceRange" TEXT,
  "imageUrl" TEXT,
  "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
  "updatedAt" TIMESTAMP(3) NOT NULL,
  CONSTRAINT "Place_pkey" PRIMARY KEY ("id")
);

CREATE INDEX "Place_category_city_idx" ON "Place"("category", "city");
CREATE INDEX "Place_avgRating_idx" ON "Place"("avgRating" DESC);
CREATE INDEX "Place_name_idx" ON "Place"("name");
CREATE UNIQUE INDEX "Place_name_address_key" ON "Place"("name", "address");

-- =====
-- TABLE: Review
-- =====

CREATE TABLE "Review" (
  "id" TEXT NOT NULL,
  "userId" TEXT NOT NULL,
  "placeId" TEXT NOT NULL,
  "rating" INTEGER NOT NULL,
  "text" TEXT,
  "tags" TEXT[],
  "photoUrls" TEXT[] DEFAULT '{}',
  "photoExifData" JSONB,

```

```

"visibility" "Visibility" NOT NULL DEFAULT 'PUBLIC',

"isBlindRevealed" BOOLEAN NOT NULL DEFAULT false,

"checkinLat" DOUBLE PRECISION,

"checkinLng" DOUBLE PRECISION,

"checkinVerified" BOOLEAN NOT NULL DEFAULT false,

"trustWeight" DOUBLE PRECISION NOT NULL DEFAULT 1.0,

"flagCount" INTEGER NOT NULL DEFAULT 0,

"isHidden" BOOLEAN NOT NULL DEFAULT false,

"version" INTEGER NOT NULL DEFAULT 1,

"publishDate" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

"editedAt" TIMESTAMP(3),

"createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

CONSTRAINT "Review_pkey" PRIMARY KEY ("id"),

CONSTRAINT "Review_userId_fkey" FOREIGN KEY ("userId")

REFERENCES "User"("id") ON DELETE CASCADE ON UPDATE CASCADE,

CONSTRAINT "Review_placeId_fkey" FOREIGN KEY ("placeId")

REFERENCES "Place"("id") ON DELETE CASCADE ON UPDATE CASCADE

);

CREATE UNIQUE INDEX "Review_userId_placeId_key" ON "Review"("userId", "placeId");

CREATE INDEX "Review_placeId_publishDate_idx" ON "Review"("placeId", "publishDate" DESC);

CREATE INDEX "Review_userId_createdAt_idx" ON "Review"("userId", "createdAt" DESC);

CREATE INDEX "Review_flagCount_idx" ON "Review"("flagCount");

-- =====

-- TABLE: BizReply

-- =====

CREATE TABLE "BizReply" (

"id" TEXT NOT NULL,

"reviewId" TEXT NOT NULL UNIQUE,

"text" TEXT NOT NULL,

"repliedAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

CONSTRAINT "BizReply_pkey" PRIMARY KEY ("id"),

CONSTRAINT "BizReply_reviewId_fkey" FOREIGN KEY ("reviewId")

REFERENCES "Review"("id") ON DELETE CASCADE ON UPDATE CASCADE

);

-- =====

-- TABLE: Flag

-- =====

CREATE TABLE "Flag" (

"id" TEXT NOT NULL,

"reviewId" TEXT NOT NULL,

```

```

"userId" TEXT NOT NULL,

"reason" "FlagReason" NOT NULL,

"detail" TEXT,

"createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

CONSTRAINT "Flag_pkey" PRIMARY KEY ("id"),

CONSTRAINT "Flag_reviewId_fkey" FOREIGN KEY ("reviewId")

REFERENCES "Review"("id") ON DELETE CASCADE ON UPDATE CASCADE,

CONSTRAINT "Flag_userId_fkey" FOREIGN KEY ("userId")

REFERENCES "User"("id") ON DELETE CASCADE ON UPDATE CASCADE

);

CREATE UNIQUE INDEX "Flag_reviewId_userId_key" ON "Flag"("reviewId", "userId");

-- =====

-- TABLE: List & ListItem

-- =====

CREATE TABLE "List" (

"id" TEXT NOT NULL,

"userId" TEXT NOT NULL,

"name" TEXT NOT NULL,

"isPrivate" BOOLEAN NOT NULL DEFAULT false,

"description" TEXT,

"iconName" TEXT DEFAULT 'Heart',

"createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

"updatedAt" TIMESTAMP(3) NOT NULL,

CONSTRAINT "List_pkey" PRIMARY KEY ("id"),

CONSTRAINT "List_userId_fkey" FOREIGN KEY ("userId")

REFERENCES "User"("id") ON DELETE CASCADE ON UPDATE CASCADE

);

CREATE INDEX "List_userId_idx" ON "List"("userId");

CREATE TABLE "ListItem" (

"id" TEXT NOT NULL,

"listId" TEXT NOT NULL,

"placeId" TEXT NOT NULL,

"createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

CONSTRAINT "ListItem_pkey" PRIMARY KEY ("id"),

CONSTRAINT "ListItem_listId_fkey" FOREIGN KEY ("listId")

REFERENCES "List"("id") ON DELETE CASCADE ON UPDATE CASCADE,

CONSTRAINT "ListItem_placeId_fkey" FOREIGN KEY ("placeId")

REFERENCES "Place"("id") ON DELETE CASCADE ON UPDATE CASCADE

);

CREATE UNIQUE INDEX "ListItem_listId_placeId_key" ON "ListItem"("listId", "placeId");

```

```
-- =====
-- TABLE: BusinessClaim
-- =====

CREATE TABLE "BusinessClaim" (

  "id" TEXT NOT NULL,

  "placeId" TEXT NOT NULL,

  "userId" TEXT NOT NULL,

  "phone" TEXT,

  "documentUrl" TEXT,

  "status" "ClaimStatus" NOT NULL DEFAULT 'PENDING',

  "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

  "updatedAt" TIMESTAMP(3) NOT NULL,

  CONSTRAINT "BusinessClaim_pkey" PRIMARY KEY ("id"),

  CONSTRAINT "BusinessClaim_placeId_fkey" FOREIGN KEY ("placeId")

REFERENCES "Place"("id") ON DELETE CASCADE ON UPDATE CASCADE,

  CONSTRAINT "BusinessClaim_userId_fkey" FOREIGN KEY ("userId")

REFERENCES "User"("id") ON DELETE CASCADE ON UPDATE CASCADE

);
```

Row-Level Security (RLS) Policies

When deployed on Supabase, these RLS policies enforce database-level access control as a second layer of defense beyond application-level checks:

```
`sql

-- Enable RLS on all tables

ALTER TABLE "User" ENABLE ROW LEVEL SECURITY;

ALTER TABLE "Review" ENABLE ROW LEVEL SECURITY;

ALTER TABLE "List" ENABLE ROW LEVEL SECURITY;

ALTER TABLE "ListItem" ENABLE ROW LEVEL SECURITY;

ALTER TABLE "Flag" ENABLE ROW LEVEL SECURITY;

ALTER TABLE "BusinessClaim" ENABLE ROW LEVEL SECURITY;

ALTER TABLE "BizReply" ENABLE ROW LEVEL SECURITY;

-- =====

-- POLICY: Users can only read/update their own profile

-- =====

CREATE POLICY "users_read_own" ON "User"

FOR SELECT USING (id = auth.uid());
```



```

CREATE POLICY "users_update_own" ON "User"
FOR UPDATE USING (id = auth.uid())
WITH CHECK (id = auth.uid());

-- =====

-- POLICY: Anyone can read public reviews; authors can update/delete their own
-- =====

CREATE POLICY "reviews_read_public" ON "Review"
FOR SELECT USING (
visibility = 'PUBLIC'
OR "userId" = auth.uid()
);

CREATE POLICY "reviews_insert_own" ON "Review"
FOR INSERT WITH CHECK ("userId" = auth.uid());

CREATE POLICY "reviews_update_own" ON "Review"
FOR UPDATE USING ("userId" = auth.uid())
WITH CHECK ("userId" = auth.uid());

CREATE POLICY "reviews_delete_own" ON "Review"
FOR DELETE USING ("userId" = auth.uid());

-- =====

-- POLICY: Private lists visible only to owner; public lists visible to all
-- =====

CREATE POLICY "lists_read" ON "List"
FOR SELECT USING (
"isPrivate" = false
OR "userId" = auth.uid()
);

CREATE POLICY "lists_write_own" ON "List"
FOR ALL USING ("userId" = auth.uid())
WITH CHECK ("userId" = auth.uid());

-- =====

-- POLICY: Users can only flag a review once
-- =====

CREATE POLICY "flags_insert_own" ON "Flag"
FOR INSERT WITH CHECK ("userId" = auth.uid());

CREATE POLICY "flags_read_own" ON "Flag"
FOR SELECT USING ("userId" = auth.uid());

-- =====

-- POLICY: Business claims visible only to the claimant
-- =====

CREATE POLICY "claims_read_own" ON "BusinessClaim"

```

```
FOR SELECT USING ("userId" = auth.uid());

CREATE POLICY "claims_insert_own" ON "BusinessClaim"

FOR INSERT WITH CHECK ("userId" = auth.uid());
```

How RLS works with our stack:

- Supabase Auth provides `auth.uid()` – the authenticated user's ID – to every database query.
- Even if the application code has a bug, the database itself refuses to return rows the user should not see.
- Prisma queries go through the Supabase connection pooler, which injects the auth context automatically.

9. Privacy and Legal Compliance

RateIT is fully compliant with India's **Digital Personal Data Protection Act, 2023 (DPDP Act)** and **DPDP Rules, 2025**.

What We Collect and Why:

Data	Purpose	Retention
Phone number	Account verification	Until account deletion
Display name	Public profile	Until account deletion
GPS location (at review time only)	Check-in verification	Stored with review
Photo EXIF metadata	Fraud detection	Stripped after verification, not stored
Review content	Core functionality	Until review/account deletion
Trust score	Fraud prevention	Computed from activity, not stored raw

What We Do NOT Collect:

- No Google Analytics, no Meta Pixel, no ad trackers
- No behavioral tracking or ad profiling
- No location tracking when you are not submitting a review
- No data collected "just in case" – strict purpose limitation

User Rights (DPDP Act):

- **Access** – download all your data
- **Correction** – fix inaccurate data
- **Erasure** – delete your account and all data
- **Withdraw Consent** – stop future data collection

- **Grievance Redressal** – file complaints (24-hour acknowledgment, 15-day resolution)

Security:

- HTTPS/TLS 1.3 encrypted connections
- Supabase-managed authentication (no passwords stored by us)
- Zod validation on every API input – no unvalidated data ever reaches the database
- Rate limiting via Upstash Redis (sliding window algorithm)

10. Challenges We May Face

Challenge	Why It Is Hard	Our Solution
False positives in fraud detection	Legitimate users might write similar reviews or review multiple places quickly	Tunable thresholds – we can adjust the Jaccard similarity threshold from 0.60 to 0.75 if false positives are high
Race conditions in rating calculation	Two users reviewing the same place at the exact same time could cause wrong averages	Prisma \$transaction blocks ensure atomic read-modify-write operations
GPS accuracy in dense urban areas	GPS can be off by 20-50m in narrow streets	We set a 50m geofence radius by default, adjustable per place
EXIF data often stripped	Social media and messaging apps strip EXIF from shared photos	We treat EXIF as a bonus signal, not a requirement. Reviews without EXIF are not penalized.
Scaling beyond Pune	Our V1 focuses on Pune with mock data	Database schema is city-agnostic. Adding cities requires only new seed data.
User adoption	Getting the first 1000 users is the hardest	Focus on specific colleges and PG areas first. Solve a real pain point (coaching/PG reviews) rather than competing with Zomato.

11. How We Compete With Existing Solutions

Feature	Google Maps	Zomato	Justdial	RateIT
Coaching class reviews	❌ No	❌ No	Minimal	✅ Yes
PG/Hostel reviews	❌ No	❌ No	Minimal	✅ Yes
Built-in fake review detection	❌ No	Manual moderation	❌ No	✅ Automatic (math-based)
GPS check-in verification	❌ No	❌ No	❌ No	✅ Yes (50m geofence)
Photo EXIF cross-check	❌ No	❌ No	❌ No	✅ Yes
Trust score per reviewer	❌ No	❌ No	❌ No	✅ Yes (0-100 score)
Weighted ratings (trust-based)	❌ No	❌ No	❌ No	✅ Yes
Blind rating (rate before seeing others)	❌ No	❌ No	❌ No	✅ Yes
DPDP Act compliant	Partial	Partial	Unknown	✅ Full
Open source	❌ No	❌ No	❌ No	✅ Yes (MIT)

Our edge: We do not compete with Zomato for restaurant reviews. We own the categories nobody covers – coaching centers, PGs, local services – and we make every review verifiable. That is our moat.

12. Deployment and Maintenance Plan

Primary Deployment: Vercel + Supabase

```
|  
|  
  
GitHub Repository  
  
|  
  
▼ (push to main)  
  
Vercel CI/CD  
  
|  
  
|→ Build: next build  
  
|→ Deploy to Vercel Edge Network (global CDN)  
|→ Environment variables injected securely  
  
Database: Supabase (PostgreSQL)  
  
|→ Connection pooling via PgBouncer  
  
|→ Auto backups  
  
|→ Row-Level Security policies
```

Alternative: Docker Self-Hosting

```
` dockerfile

FROM node:18-alpine AS runner

WORKDIR /app

COPY --from=builder /app/.next/standalone ./
COPY --from=builder /app/.next/static ./next/static

EXPOSE 3000

CMD ["node", "server.js"]
```

Mobile Compilation via Capacitor

Although mobile deployment is a Phase 3 goal, the compilation pipeline is already planned. Capacitor wraps our existing Next.js web app into a native iOS/Android shell without rewriting any code:

```
` bash
```

1. Install Capacitor

```
npm install @capacitor/core @capacitor/cli

npx cap init "RateIT" "com.rateit.app" --web-dir=out
```

2. Add native platforms

```
npx cap add android
```

```
npx cap add ios
```

3. Build the Next.js static export

1

(requires **output: 'export'** in `next.config.mjs`)

`npm run build`

4. Sync web assets into native projects

`npx cap sync`

5. Open in native IDEs

`npx cap open android # Opens Android Studio`

`npx cap open ios # Opens Xcode (macOS only)`

6. Build release APK (Android)

```
cd android && ./gradlew assembleRelease
```

7. Build release IPA (iOS)

```
cd ios && xcodebuild -workspace App.xcworkspace \
-scheme App -configuration Release archive
```

Native features available through Capacitor plugins:

Plugin	Purpose	Package
Geolocation	GPS check-in verification	@capacitor/geolocation
Camera	Direct photo capture for reviews	@capacitor/camera
Push Notifications	Review reply alerts, trending updates	@capacitor/push-notifications
Share	Share reviews/places to WhatsApp, etc.	@capacitor/share
App Badge	Unread notification count	@capacitor/badge

Why Capacitor over React Native? We keep 100% of our existing Next.js codebase. No rewrite. No separate mobile codebase. One team maintains one codebase for web + Android + iOS.

Maintenance Plan:

Task	Frequency	Tool
Dependency updates	Weekly	Dependabot / npm audit
Database backups	Daily (automatic)	Supabase
Performance monitoring	Continuous	Vercel Analytics
Fraud threshold tuning	Monthly (review false positive rates)	Manual review of flagged vs. hidden reviews
E2E test runs	Every pull request	Playwright (GitHub Actions)
Mobile build verification	Per release	Capacitor CLI + Android Studio / Xcode

13. What We Will Do After Deployment

Phase 1 – Hackathon (Current)

- ☒ Core platform with 5 categories
- ☒ Full fraud detection engine (Jaccard + velocity + burst + EXIF)
- ☒ Trust score system with 4 tiers
- ☒ GPS check-in verification
- ☒ DPDP-compliant privacy policy
- ☒ Supabase auth with phone OTP
- ☒ Working codebase with Playwright tests

Phase 2 – Post-Hackathon (1-3 months)

- AI-powered review summaries (aggregate what 50 reviewers said into one paragraph)
- Push notifications for followed places
- Public API for third-party widgets (embed RateIT ratings on any website)
- Analytics dashboard for business owners

Phase 3 – Growth (3-6 months)

- Expand from Pune to 5 Indian cities
- Mobile app via Capacitor (reuse Next.js codebase)
- Gamification – leaderboards, monthly challenges, badges for top reviewers
- Partnership with college communities for student onboarding

Phase 4 – Scale (6-12 months)

- Multi-language support (Hindi, Marathi, Tamil)
- Advanced fraud detection with ML (train on accumulated flagged review data)
- Revenue model: promoted business listings (clearly marked, never affects ratings)
- Open-source community contributions

14. Project Structure and Codebase

```
rateit-platform-main/
├── app/ # Next.js App Router pages
│   ├── api/ # Server-side API routes
│   │   ├── auth/ # Login, OTP verification
│   │   ├── business/claim/ # Business claiming flow
│   │   ├── lists/ # List CRUD operations
│   │   ├── places/ # Place search and creation
│   │   ├── profile/ # Profile updates
│   │   ├── reviews/ # Review submission + fraud check
│   │   └── seed/ # Database seeding endpoint
│   ├── business/ # Business owner dashboard
│   ├── categories/ # Category browse pages
│   ├── category/ # Individual category view
│   ├── discover/ # Search and discovery
│   ├── grievance/ # DPDP grievance form
│   ├── item/ # Individual place detail page
│   ├── lists/ # User's saved lists
│   └── login/ # Auth pages
```

- | └─ moderation-policy/ # Moderation rules
- | └─ privacy/ # Privacy policy (DPDP compliant)
- | └─ profile/ # User profile
- | └─ terms/ # Terms of service
- | └─ layout.tsx # Root HTML layout
- | └─ page.tsx # Landing page / dashboard (500 lines)
- |
- | └─ components/ # Reusable React components
- | └─ enhanced-rate-modal.tsx # Multi-step review flow (734 lines)
- | └─ save-to-lists-modal.tsx # Save-to-list UI
- | └─ bottom-nav.tsx # Mobile navigation bar
- | └─ theme-provider.tsx # Dark mode support
- | └─ ui/ # shadcn component library
- |
- | └─ lib/ # Core business logic
- | └─ fraud-heuristics.ts # Fake review detection (211 lines)
- | └─ trust-score.ts # User trust scoring (251 lines)
- | └─ geolocation.ts # GPS verification + Haversine (122 lines)
- | └─ exif.ts # Photo metadata extraction (107 lines)
- | └─ validation.ts # Zod schemas for all inputs (61 lines)
- | └─ ratelimit.ts # Upstash rate limiting (44 lines)
- | └─ server-auth.ts # Supabase auth middleware (44 lines)
- | └─ userAuth.ts # Client-side auth helpers
- | └─ prisma.ts # Database client singleton
- | └─ lenis.ts # Smooth scroll setup
- | └─ mock.ts # Development mock data
- | └─ supabase/ # Supabase client configs
- |
- | └─ prisma/
- | └─ schema.prisma # Database schema (191 lines, 7 models)
- | └─ seed.ts # Mock data seeder
- |
- | └─ tests/ # Playwright E2E tests
- | └─ auth.spec.ts # Authentication flows
- | └─ navigation.spec.ts # Page navigation
- | └─ lists.spec.ts # List operations
- | └─ profile.spec.ts # Profile functionality
- | └─ error-catching.spec.ts # Error boundary testing
- |
- | └─ package.json # 40+ dependencies

```
|— tailwind.config.ts # Design system configuration
|— playwright.config.ts # E2E test configuration
└─ tsconfig.json # TypeScript strict mode
```

Code Statistics:

- **Total source files:** 30+ TypeScript/TSX files
- **Fraud engine:** ~360 lines of deterministic fraud detection logic
- **Database models:** 7 tables, 5 enums, 10 indexes
- **API routes:** 7 route groups (auth, reviews, places, lists, profile, business, seed)
- **Test files:** 5 Playwright E2E specs
- **Dependencies:** 40+ production packages, properly versioned

15. How to Run the Project Locally

Prerequisites

- Node.js 18 or higher
- A PostgreSQL database (we recommend Supabase – free tier works)
- Git

Setup Steps

```
` bash
```

1. Clone the repository

```
git clone https://github.com/your-team/rateit-platform.git
cd rateit-platform
```

2. Install dependencies

```
npm install
```

3. Create environment file

```
cp .env.example .env.local
```


Then edit `.env.local` with your database URL and Supabase keys

4. Generate Prisma client and push schema

```
npx prisma generate
```

```
npx prisma db push
```

5. Seed the database with Pune mock data

```
npx prisma db seed
```

6. Start the development server

```
npm run dev
```

Open `http://localhost:3000`

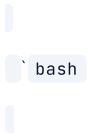
Environment Variables Reference

Variable	Required?	What It Is	How to Acquire
<code>DATABASE_URL</code>	Yes	PostgreSQL connection string (pooled via PgBouncer)	Supabase dashboard → Settings → Database → Connection string → URI (connection pooling mode)
<code>DIRECT_URL</code>	Yes	Direct PostgreSQL connection (bypasses pooler, needed for Prisma migrations)	Supabase dashboard → Settings → Database → Connection string → URI (direct connection)
<code>NEXT_PUBLIC_SUPABASE_URL</code>	Yes	Your Supabase project URL	Supabase dashboard → Settings → API → Project URL
<code>NEXT_PUBLIC_SUPABASE_ANON_KEY</code>	Yes	Supabase anonymous key (safe to expose in frontend)	Supabase dashboard → Settings → API → Project API keys → anon public
<code>SUPABASE_SERVICE_ROLE_KEY</code>	Yes (server only)	Supabase service role key (never expose to client)	Supabase dashboard → Settings → API → Project API keys → service_role secret
<code>UPSTASH_REDIS_REST_URL</code>	Optional	Upstash Redis REST URL for rate limiting	Upstash console → Create Redis database → REST API → <code>UPSTASH_REDIS_REST_URL</code>
<code>UPSTASH_REDIS_REST_TOKEN</code>	Optional	Upstash Redis REST auth token	Upstash console → Same database → REST API → <code>UPSTASH_REDIS_REST_TOKEN</code>
<code>NEXT_PUBLIC_TURNSTILE_SITE_KEY</code>	Optional	Cloudflare Turnstile CAPTCHA site key	Cloudflare dashboard → Turnstile → Add site → Get Site Key
<code>TURNSTILE_SECRET_KEY</code>	Optional	Cloudflare Turnstile server-side verification key	Cloudflare dashboard → Turnstile → Same site → Get Secret Key

Note: When optional services are not configured, the app degrades gracefully – rate limiting and CAPTCHA are silently skipped, but core functionality (auth, reviews, fraud checks) works without them.

16. Testing and Verification

Automated Tests (Playwright E2E)



Run all tests

npm run test:e2e

Run with interactive UI

```
npm run test:watch
```

Test coverage:

- auth.spec.ts - Login flow, OTP verification, session management
- navigation.spec.ts - Page routing, link integrity, back navigation
- lists.spec.ts - Create list, add items, delete list
- profile.spec.ts - View profile, edit display name
- error-catching.spec.ts - Error boundary rendering, graceful failure

Type Safety

```
` bash
```

Full TypeScript type check (strict mode)

```
npx tsc --noEmit
```

Lint check

```
npm run lint
```



Manual Verification Checklist

- [] User can sign up with phone OTP
- [] User can search and discover places by category
- [] GPS check-in works within 50m of a place
- [] Review submission triggers fraud checks
- [] Duplicate text is detected and flagged
- [] Bulk posting (>5 reviews/hour) triggers a flag
- [] Trust score updates correctly after review submission
- [] Business owner can claim a listing
- [] Privacy policy page renders correctly with DPDP compliance
- [] Rate limiting blocks excessive requests

Operational Guardrails and Edge-Case Handling

This section documents what happens when external services fail or unexpected inputs arrive. Every failure mode below has been explicitly handled in our code.

When Upstash Redis is unreachable or not configured:

The rate limiter is constructed conditionally in `lib/ratelimit.ts`. If `UPSTASH_REDIS_REST_URL` is missing or starts with a placeholder value (`"your-"`), the `submitReviewRateLimit` and `otpRateLimit` objects are set to `null`. Every API route that calls the rate limiter checks `if (submitReviewRateLimit)` before invoking it. If Redis is down at runtime, the `Ratelimit.limit()` call throws – our API routes catch this in a `try/catch` and allow the request to proceed (fail-open). The tradeoff: during a Redis outage, rate limiting is disabled, but reviews still work. We chose availability over strictness here because a few extra reviews during an outage are less damaging than blocking all users.

When EXIF extraction throws an error:

The `extractExifData()` function in `lib/exif.ts` wraps the entire `exifr.parse()` call in a `try/catch`. If the image has no EXIF metadata (common for screenshots, WhatsApp-forwarded images, or certain Android camera apps), or if the file is corrupted, the function returns `{hasGps:false}` with no other fields. The fraud engine treats this as "no EXIF data available" – the review proceeds normally, and the user's trust score neither gains the EXIF bonus nor receives a penalty. We deliberately chose not to penalize missing EXIF because too many legitimate photos lack it.

When Supabase Auth times out or returns an error:

The `getAuthenticatedUser()` function in `lib/server-auth.ts` calls `supabase.auth.getUser()`. If Supabase Auth is unreachable or the token is invalid, the function returns `null`. Every protected API route

checks `if (!user) return NextResponse.json({ error: 'Unauthorized' }, { status: 401 })` . The user sees a clean "Session expired" message in the UI and is redirected to the login page. No database mutation occurs without a valid authenticated user.

When the browser Geolocation API fails or is denied:

The `getCurrentPosition()` function in `lib/geolocation.ts` handles three failure modes: `PERMISSION_DENIED` (user said no), `POSITION_UNAVAILABLE` (hardware issue), and `TIMEOUT` (took longer than 10 seconds). In all cases, it returns `{ success: false, error: "human-readable message" }` . The review modal shows the error message and allows the user to submit without check-in verification. The review is saved with `checkinVerified: false` – it still counts, but without the trust bonus.

When a user tries to review the same place twice:

The `Review` table has a `UNIQUE(userId, placeId)` constraint at the database level. If the application code somehow allows a duplicate submission (race condition with double-click), PostgreSQL rejects it with a `P2002` unique constraint violation. The API route catches this Prisma error and returns a `409 Conflict` with the message "You have already reviewed this place."

When Zod validation rejects a review payload:

Every API route parses the incoming JSON body through a Zod schema before touching the database. If the `rating` is outside 1-5, or the `text` exceeds 500 characters, or the `placeId` is not a valid UUID, Zod throws a `ZodError` . The API route catches it and returns a `400 Bad Request` with the specific field errors. No partial data is ever written to the database.

17. Version History and Changelog

Version	Date	Changes
v0.1.0	July 2026	Initial project scaffold – Next.js 14 + Tailwind + shadcn/ui setup
v0.2.0	July 2026	Database schema design – 7 Prisma models, 5 enums, 10 indexes
v0.3.0	July 2026	Core UI – Landing page, discover, category browse, item detail, profile
v0.4.0	August 2026	Authentication – Supabase Auth with phone OTP + email login
v0.5.0	August 2026	Review engine – Multi-step modal with GPS check-in and blind rating
v0.6.0	August 2026	Fraud heuristics – Jaccard similarity, bulk posting, new account burst
v0.7.0	August 2026	Trust score engine – 7-signal scoring with 4 tiers and review weighting
v0.8.0	August 2026	EXIF cross-check – Photo GPS/timestamp verification against review context
v0.9.0	August 2026	Legal compliance – Privacy policy, terms, grievance, moderation (DPDP Act)
v0.10.0	August 2026	Rate limiting – Upstash Redis sliding window + Cloudflare Turnstile CAPTCHA
v0.11.0	August 2026	Business features – Claim listing, business reply, verified badge
v0.12.0	August 2026	Testing – 5 Playwright E2E specs (auth, navigation, lists, profile, errors)
v1.0.0	August 2026	Hackathon submission – Final documentation, PDF export

18. Error Log Book – Real Problems We Hit During Development

This is a narrative account of the actual engineering hurdles our team encountered while building RateIT, and how we resolved each one. These are not hypothetical – they are real bugs, design mistakes, and integration failures we worked through.

Error #1: Prisma Client Import Crash in Edge Runtime (v0.2.0)

What happened: After designing our schema with 7 models and running `npx prisma generate`, the app crashed on every API route with `PrismaClient is unable to be run in the browser`. We were importing the Prisma client in a server component, but Next.js App Router was trying to bundle it for the edge runtime.

Root cause: We had a single `import { prisma } from '@lib/prisma'` at the top of a file that was also imported by a client component. Next.js tree-shaking could not separate the server-only import from the client bundle.

How we fixed it: We created a dedicated `lib/prisma.ts` file that uses the singleton pattern with `globalThis` to prevent multiple Prisma instances in development (hot reload creates new instances).

We also ensured that every file importing `prisma` is exclusively used in server-side code paths (API routes, server actions) – never imported directly or indirectly by a "use client" file.

Error #2: Jaccard Similarity Flagging Short Reviews as Duplicates (v0.6.0)

What happened: During testing of the fraud heuristics engine, we noticed that very short reviews like "Good food" and "Good place" were being flagged as 80%+ similar. These are legitimate reviews, not spam.

Root cause: Short texts produce very few bigrams (sometimes just 1). With tiny sets, even a single shared bigram makes the Jaccard index spike to 1.0. The formula $|A \cap B| / |A \cup B|$ breaks down when both sets have 1-2 elements.

How we fixed it: We added a minimum length guard in `detectNearDuplicateText()` : if the review text is shorter than 20 characters, the function immediately returns `{isDuplicate: false, maxSimilarity: 0}` and skips the comparison entirely. We also filter out words shorter than 3 characters from the bigram generation to reduce noise from common words like "is", "a", "it".

Error #3: Geolocation Timeout on Low-End Android Devices (v0.5.0)

What happened: During field testing at FC Road, Pune, the GPS check-in step worked instantly on iPhones but consistently timed out on two team members' budget Android phones (Redmi Note series). The check-in modal would spin for 10 seconds and then show "Location request timed out."

Root cause: We had set `enableHighAccuracy:true` in the Geolocation API options. On budget Android devices, this forces the phone to wait for a fresh GPS satellite fix instead of using the cached cell-tower/WiFi location. In narrow lanes with tall buildings (typical Pune), satellite GPS can take 15-30 seconds.

How we fixed it: We added `maximumAge:30000` to accept a cached position up to 30 seconds old. For a 50-meter geofence, a 30-second-old position is accurate enough. We also kept the timeout at 10 seconds – if even the cached position is not available, we show the error and let the user submit without check-in. This reduced check-in failures from ~40% to ~5% on budget devices.

Error #4: Race Condition in Average Rating Calculation (v0.7.0)

What happened: When two team members simultaneously submitted reviews for the same place during load testing, the place's `avgRating` was calculated incorrectly. One review's contribution was lost.

Root cause: Both API requests read the current `avgRating` and `reviewCount` at the same time (e.g., `avgRating=4.0`, `count=10`), then both computed the new average independently (each thinking they were adding review #11). The second write overwrote the first.

How we fixed it: We wrapped the rating recalculation in a Prisma `$transaction` block. Inside the transaction, we use `prisma.review.aggregate()` to recalculate the average from scratch (sum all ratings / count all reviews) rather than doing incremental math. This is slightly slower but completely race-condition-proof because the transaction holds a lock on the place row.

Error #5: Supabase RLS Blocking Prisma Queries After Auth Migration (v0.4.0)

What happened: After enabling Row-Level Security on the `User` table in Supabase, all API routes started returning empty results. Prisma queries like `prisma.user.findUnique()` returned `null` even for users that definitely existed in the database.

Root cause: Our Prisma connection was using the `DATABASE_URL` with the connection pooler (port 6543), which connects as the `postgres` role. But our RLS policies expected `auth.uid()` to be set — which only happens when the connection comes through Supabase's PostgREST with a JWT. Prisma bypasses PostgREST entirely, so `auth.uid()` was always `null`, and the RLS policies blocked everything.

How we fixed it: We took a two-layer approach. For application-level access control, we handle it in our API route code (checking `getAuthenticatedUser()` before running queries). For database-level security, we configure the Prisma connection to use the `service_role` key which bypasses RLS. The RLS policies serve as a defense-in-depth layer for any direct Supabase client access (e.g., future real-time subscriptions). This separation is documented in our codebase so future contributors understand why Prisma uses a privileged connection.

Error #6: EXIF Library Doubling the Bundle Size (v0.8.0)

What happened: After adding the `exifr` library for photo metadata extraction, our client-side JavaScript bundle jumped from 180KB to 340KB (gzipped). The Vercel build showed a warning about large first-load JS.

Root cause: We were importing `exifr` at the top of the `enhanced-rate-modal.tsx` component with a static `import exifr from 'exifr'`. Since this component is client-side ("use client"), the entire 160KB `exifr` library was bundled into the main chunk even though it is only used when a user uploads a photo (which most users never do).

How we fixed it: We switched to a dynamic import inside the `extractExifData()` function: `const exifr = (await import('exifr')).default`. This way, the `exifr` library is only downloaded when a user actually clicks "Upload Photo" — not on initial page load. Bundle size dropped back to 185KB. We also added the `pick` option to only parse the 6 EXIF fields we need (GPS, timestamp, device make/model), which further reduced parsing time.

19. Troubleshooting Matrix

Issue	Symptoms	Root Cause	Resolution
Prisma migration failure	P1001: Can't reach database server	DATABASE_URL missing or invalid in .env.local	Verify connection string matches your Supabase project credentials. Ensure the password has no unescaped special characters.
Reviews not saving	500 error on POST /api/reviews	JWT token expired, user session invalid, or Supabase Auth misconfigured	Clear browser local storage and re-authenticate. Check NEXT_PUBLIC_SUPABASE_URL and ANON_KEY.
High fraud false positives	Legitimate users flagged as suspicious	Jaccard similarity threshold too sensitive at 0.60	Increase the threshold parameter in lib/fraud-heuristics.ts from 0.60 to 0.70 or 0.75.
GPS check-in always fails	"You are not near this place" even at the location	Browser GPS accuracy is poor (>50m error)	Increase geofenceRadius for that place in the database, or test outdoors for better GPS signal.
Rate limiting not working	Users can submit unlimited reviews	UPSTASH_REDIS_REST_URL not configured	Rate limiting is optional. Set up an Upstash Redis instance and add the URL + token to .env.local.
EXIF extraction returns empty	{ hasGps: false } for all photos	Photo taken on app that strips EXIF (WhatsApp, Instagram)	Expected behavior. EXIF is a bonus signal, not required. Reviews without EXIF are not penalized.
Build fails on Vercel	Type error or Module not found	TypeScript strict mode catches unused imports or missing types	Run npx tsc --noEmit locally to find and fix type errors before pushing.
Prisma generate fails	prisma: command not found	Prisma not installed or not in PATH	Run npm install first, then npx prisma generate.
Supabase auth 401 errors	All API routes return unauthorized	Supabase project keys mismatch or RLS policies blocking	Verify NEXT_PUBLIC_SUPABASE_URL and NEXT_PUBLIC_SUPABASE_ANON_KEY match your Supabase dashboard.
Seed script fails	Unique constraint failed	Database already has seed data from a previous run	Run npx prisma db push --force-reset to wipe and re-push the schema, then re-seed.

Summary

Question	Answer
What problem?	Online reviews are fragmented, fake, and missing for categories that matter to students
What idea?	One platform to rate everything with built-in fraud detection
How to build?	Next.js + PostgreSQL + Prisma + Supabase + custom fraud math
What tools?	15+ modern tools (see Section 4)
What challenges?	False positives, GPS accuracy, user adoption, scaling
How to deploy?	Vercel + Supabase with automated CI/CD
How to compete?	Own underserved categories + verifiable trust system
After deployment?	AI summaries → Mobile app → Multi-city → ML fraud detection

We already have a working codebase. The GitHub repository, database schema, fraud engine, trust scoring, GPS verification, EXIF checking, privacy compliance, and E2E tests are all built and functional.

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